Received: Added at production | Revised: Added at production | Accepted: Added at production

DOI: xxx/xxxx

SPECIAL ISSUE ARTICLE

CAE-WANN: A Novel Anomaly Detection Method For Gas Turbines Via Search Space Extension†

Shisheng Zhong*[1] | Dan Liu[1] | Lin Lin[1] | Minghang Zhao*[2] | Xuyun Fu[2] | Feng Guo[1]

[1]School of Mechatronics Engineering, Harbin Institute of Technology, Harbin, China

[2]School of Ocean Engineering, Harbin Institute of Technology at Weihai, Weihai, China

Correspondence
Shisheng Zhong, School of Mechatronics Engineering, Harbin Institute of Technology, 150001, Harbin, China. Email: zhongss@hit.edu.cn
Minghang Zhao, School of Ocean Engineering, Harbin Institute of Technology at Weihai, 264209, Weihai, China. Email: zhaomh@hit.edu.cn

Summary

Anomaly detection of gas turbines is a typical binary classification problem under small sample size. In essence, the goal of classification is to discover the mapping that can map samples in different categories to disjoint codomains. The neural network has strong nonlinear mapping ability thus it has great potentialities in classification tasks. However, how to design the architecture of the neural network is still a challenging problem. The inappropriate network architecture might cause either overfitting or underfitting problems. To alleviate the problem, this paper presents an effective anomaly detection model for gas turbines by combining the convolutional auto-encoder and weight agnostic neural network search. More specifically, the convolutional auto-encoder is used for the search space extension by obtaining high-dimensional latent representations of the raw data. Then, weight agnostic neural network search finds a suitable neural network architecture for anomaly detection in the extended search space. Moreover, a novel method based on maximum likelihood estimation is proposed for threshold selection, which is also effective in the case of unbalanced datasets. In the end, the real-life monitoring data of gas turbines validate the effectiveness of the presented anomaly detection model.

KEYWORDS:
gas turbines, anomaly detection, small samples, convolutional auto-encoders, weight agnostic neural network search

1 | INTRODUCTION

As one of the most important parts of an airplane, the gas turbine has a decisive influence on the safety of passengers during a flight. If the abnormality can be detected before a flight accident, airlines can repair or replace the components or gas turbine to avoid major field failures as soon as possible. Besides, it can also reduce operating costs and aircraft downtime to avoid significant economic losses.[1,2,3,4] Therefore, it is necessary to carry out real-time condition monitoring and anomaly detection of the gas turbine.

The popular anomaly detection methods for gas turbines mainly include (1) model-based[5], (2) statistics-based[6], and (3) machine learning-based. The model-based method requires developing accurate and comprehensive physical models for gas turbines, which could be very arduous because the damage propagation processes and dynamic responses of complex equipment are very complicated[7]. The statistics-based method handles variations in the statistical property of the sensor data using a customizable probability model. To devise the probability model, it requires prior knowledge of gas turbines, which is very difficult

†National Natural Science Foundation of China (No.U1733201)

to obtain[8]. However, the machine learning method can extract features from real-life monitoring data without understanding the operating mechanism of gas turbines[9]. As an important branch of machine learning, artificial neural networks(ANNs) have attracted increasing attention in the anomaly detection and fault diagnosis for gas turbines in recent years[1 10 11 12 13 14]. For example, Fu et al.[10] employed a stacked denoising autoencoder(SDAE) to extract features from raw data for yielding high detection accuracy. Yuan et al.[14] applied a Long Short-Term Memory(LSTM) neural network for fault diagnosis of aero-engine, which got a good diagnosis in the case of strong noises.

However, how to design the architecture of the neural network is still a challenging problem. On one hand, if the network architecture is extremely simple, it is unable to solve complex tasks in practical engineering applications owing to underfitting. On the other hand, if the network architecture is highly complex, it exhibits a poor generalization ability owing to overfitting. Even the more popular convolutional neural networks(CNNs), recurrent neural networks(RNNs), and many hybrid models with attention mechanisms have this problem as well[15 16 17]. In addition, it is hard to collect large amounts of abnormal data from an airline fleet, because gas turbines have a low failure frequency[18]. The problem becomes even worse under small sample size[1].

To alleviate this problem, search methods for network architectures have raised more attention and achieved success in many complex tasks[19 20]. The weight agnostic neural network search method is developed in view of the problem that previous neural architecture search algorithms are inefficient[21]. Moreover, an objective function is introduced for avoiding overfitting or underfitting by combining three-loss values. On one hand, the mean and max performance loss are used to minimize the error between the sigmoid output of the neural network and the corresponding label for avoiding underfitting. On the other hand, complexity loss is used to reduce the number of connections in neural networks for avoiding overfitting. Therefore, it is interesting to apply it to anomaly detection of gas turbines[1].

Search space defines the set of all network architectures that can be found theoretically using network architecture search methods and network search represents the process that finds the well-performing network architecture quickly in search space[22]. The number of network architectures in search space is determined by the number of input nodes for weight agnostic neural networks(WANNs). If the number of input nodes is relatively small, the number of network architectures in the corresponding search space will be relatively small, which will have a certain negative impact on WANNs[22]. In principle, better network architecture can be searched via search space extension, that is, increasing the number of network architectures in the search space. Therefore, it is necessary to explore a universal search space extension method to increase the performance of WANNs. In this paper, on the premise of keeping the main and important characteristics of the raw data as much as possible, the raw data is mapped from low-dimensional space to high-dimensional space using linear or nonlinear mapping, and then the high-dimensional latent representation is taken as the input of the WANN to achieve the purpose of search space extension.

Many low-dimensional and high-dimensional spatial mapping methods are based on the encoder-decoder framework[23], such as sparse auto-encoders(SAEs), stacked sparse auto-encoders(SSAEs), and so on. The raw data is first mapped to a high-dimensional space (encoder) and then back to the original space (decoder). After the model training is completed, the high-dimensional latent representation of the raw data can be obtained. In fact, the raw data is the two-dimensional grid data, which is shown in TABLE 1. However, both fully-connected SAE and SSAE ignore the structure of the two-dimensional grid data[24]. Fortunately, convolutional auto-encoders(CAEs) can solve this problem well by utilizing the convolution kernels of CNNs[25]. Inspired by the above research, this paper presents a new gas turbine anomaly detection model by combining the CAE and WANN. First of all, the WANN performs remarkably well even in the high-dimensional classification task[21]. The WANN has the significant potential for fitting the real-life monitoring data of gas turbines and distinguishing abnormal samples from normal samples. Secondly, the CAE is used to obtain high-dimensional latent representations of the raw data. And the high-dimensional latent representations are more easily distinguished by the WANN than the raw data, which can effectively reduce the difficulty of the diagnosis. Finally, the complexity of the neural network structure is considered as a term in the objective function, which can significantly reduce the number of learnable parameters, avoiding the risk of overfitting even under the condition of small observation samples[1]. Therefore, the proposed method is applicable to small sample size data. Moreover, the main contributions of this paper are as follows:

- This paper proposes a novel supervised anomaly detection framework by integrating the CAE with the WANN, which effectively improves the anomaly detection performance for gas turbines under small sample size.
- This paper utilizes the CAE to obtain easily distinguishable high-dimensional latent representations of the raw data, which significantly increases the performance of WANN.
- This paper uses maximum likelihood estimation to set a proper threshold, which can also distinguish abnormal samples from normal samples in case of unbalanced datasets.

The rest of this paper is organized as follows. In the first place, Section 2 describes the construction process of the CAE-WANN, introduces the weight agnostic neural network search and convolutional auto-encoders, and analyses the threshold selection under unbalanced samples. Additionally, Section 3 gives a practical example that validates the effectiveness of the presented anomaly detection model. Finally, Section 4 discusses the conclusion and future work.

2 | THEORY OF THE DEVELOPED CAE-WANN

As stated in Section 1, the integration of a CAE and a WANN is considered in this paper. Hence, this section focuses on developing the CAE-WANN. But before this, it is necessary to briefly introduce the search space of weight agnostic neural network search.

2.1 | Search Space

The partition schematic diagram of the WANN is shown in FIGURE 1. Nodes in different layers are marked in different colors. The connection between layers in WANN is sparse and skip connection, as can be seen from the figure. Therefore, its structure is very different from a backpropagation neural network. To illustrate the effect of input nodes on the number of network architectures in search space, the nodes in the WANN can be divided into two disjoint subsets. If W represents the set of all nodes, U represents the set of input nodes, V represents the set of hidden nodes and output nodes, then $W = U \cup V$, and $U \cap V = \varnothing$.

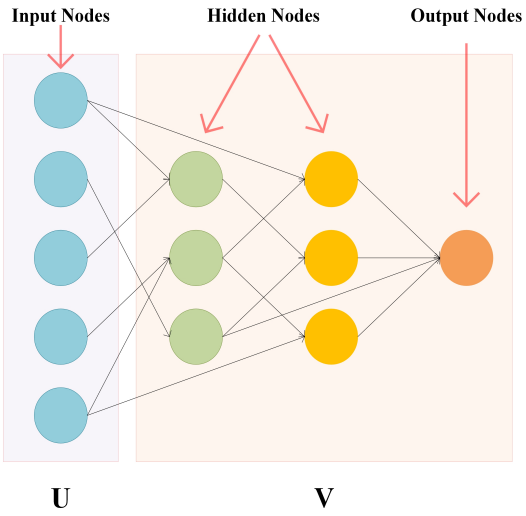


FIGURE 1 The partition of the WANN

Unbounded search space has no practical significance, so the search space needs to be constrained to have a finite (potentially very large) number of nodes[22]. Now assume that U contains m nodes, V contains n nodes, and N_V is the number of network architectures composed of nodes in V. Obviously, N_V grows exponentially with n. In mathematics, the directed acyclic graph is a directed graph with no directed cycles. The number of directed acyclic graphs is a bounded constant[26]. As stated in the above definition, the WANN is a directed acyclic graph. Because the number of output nodes was fixed in this paper, the directed acyclic graph may not be a WANN. Therefore, the number of network architectures is not greater than that of directed acyclic graphs. That is, N_V is also a bounded constant.

Theorem 1. If $U = \{u_1, u_2, \cdots, u_m\}$,$V = \{v_1, v_2, \cdots, v_n\}$, then the number of network architectures in search space is

$$N = (2^n \cdot N_V)^m \tag{1}$$

Proof. If $m = 1$, then $U = \{u_1\}$. For any node v_j in V, there are only two connection states between u_1 and v_j: connection and non-connection. Since V contains n nodes, there are 2^n connection states between u_1 and V. Note that N_V is the number of network architectures composed of nodes in V. Then $N = 2^n \cdot N_V = (2^n \cdot N_V)^1$. So it is true when $m = 1$.

Now assume that $N = (2^n \cdot N_V)^m$ is true when $m = k$. Since there is no connection between input nodes, $U = \{u_1, u_2, \cdots, u_{k+1}\}$ can be divided into two disjoint subsets $U_1 = \{u_1, u_2, \cdots, u_k\}$ and $U_2 = \{u_{k+1}\}$ when $m = k + 1$. Because U_1 contains k nodes and satisfies the hypothesis $m = k$, there are $(2^n \cdot N_V)^k$ connection states between U_1 and V. And U_2 only contains one node and satisfies the hypothesis $m = 1$, so there are $2^n \cdot N_V$ connection states between U_2 and V. Then $N = (2^n \cdot N_V) \cdot (2^n \cdot N_V)^k = (2^n \cdot N_V)^{k+1}$. So it is true when $m = k + 1$. In other words, Theorem 1 is proved.

□

According to Theorem 1, the number of network architectures in search space increases exponentially with the number of input nodes. Therefore, increasing the number of input nodes can greatly increase the number of network architectures in search space, and better network architectures can be searched in principle.

2.2 | Design of Fundamental Architectures for the CAE-WANN

The overall framework of the developed CAE-WANN is shown in FIGURE 2, which mainly includes two components: a CAE for obtaining high-dimensional latent representations of the raw data and a WANN for diagnosing the abnormality of the gas turbine.

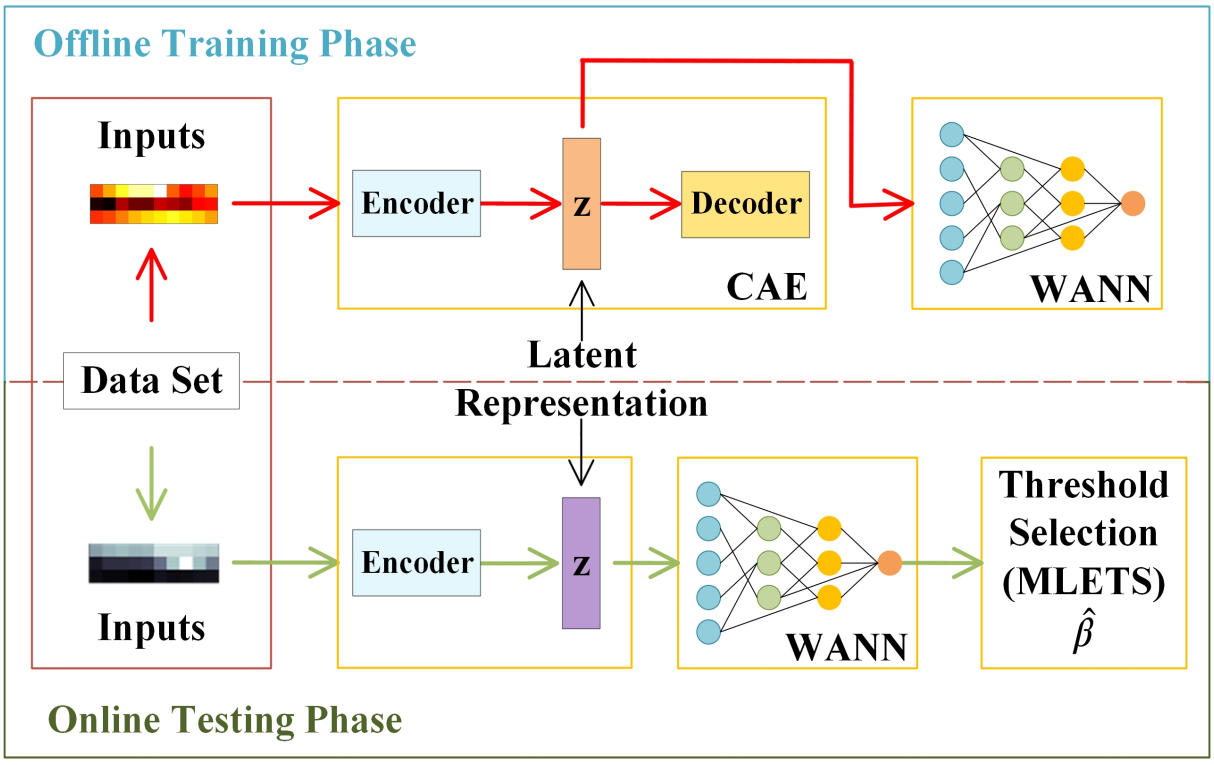


FIGURE 2 The overall framework of the developed anomaly detection model

After collecting the real-life monitoring data of gas turbines, it is necessary to train the CAE and WANN. The common training procedure is as follows(FIGURE 3 (a)): (1) The CAE is trained with the raw data, and the labels are the reconstructed representation of the raw data. The CAE is updated by the reconstruction loss until it is well-trained. In this step, the WANN is not updated. (2) The high-dimensional latent representation of the raw data is obtained with the well-trained encoder. (3) The WANN is trained with the high-dimensional latent representation, and the labels are zero or one, representing the normal or abnormal state of the gas turbine. The WANN is updated by the objective function until it is well-trained. In this step, the

CAE is not updated. Since the CAE in this paper is used to increase the dimension of the raw data, there is often more than one high-dimensional latent representation of the raw data. To obtain the high-dimensional latent representation that is easily distinguished by WANNs, this paper presents an alternating training approach that the CAE-WANN is trained by updating the CAE and WANN alternatively. The specific steps are as follows(FIGURE 3 (b)): (1) The CAE is trained with the raw data. The CAE is updated once by the feedback loss. (2) The high-dimensional latent representation of the raw data is obtained with the encoder. (3) The WANN is trained with the high-dimensional latent representation. The WANN is updated once by the objective function. Then, repeat (1) to (3) until a suitable network architecture is searched. And each component of the proposed model will be introduced separately in the next step.

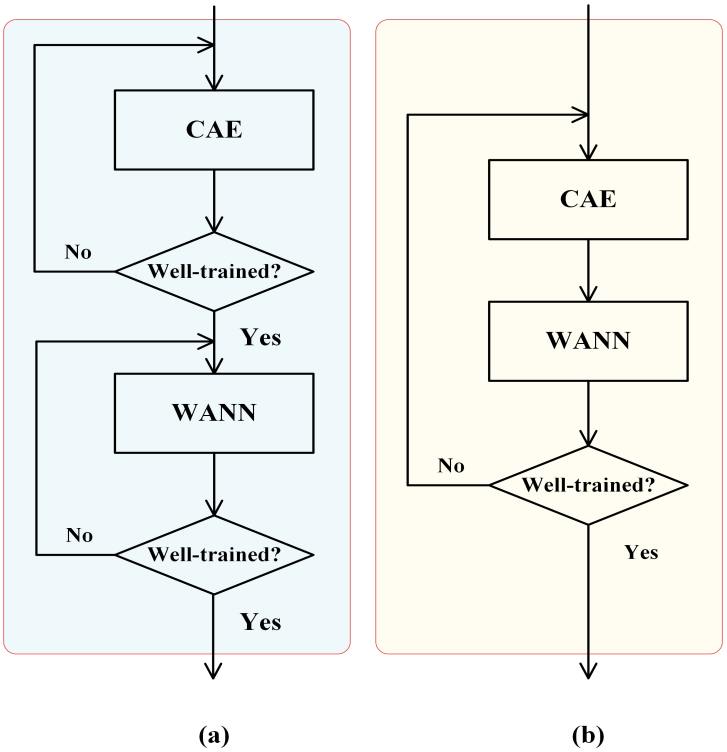


FIGURE 3 The training procedure of the CAE-WANN

2.3 | Convolutional Auto-Encoder

The CAE has properties of CNNs and can directly process the grid data. As an important variant of the AE, the CAE is also composed of an encoder and a decoder. In the first place, the encoder maps the input x to a high-dimensional space, that is, $z = Encoder(x)$. In contrast to the encoder, the decoder maps the latent representation z back to the original space, that is, $\hat{x} = Decoder(z)$.

To improve training stability, in addition, the architectural topology of the CAE in this paper is adjusted according to the constraints of the DCGAN[27]. The encoder consists of convolutional layers, batch normalization, and leaky ReLU activation, while the decoder consists of deconvolutional transpose layers, batch normalization, and ReLU activation together with a tanh layer at the end, as can be seen from FIGURE 4. Moreover, these architecture constraints for stable CAE are as follows:

- Replace any pooling layers with strided convolutions(Encoder) and fractional-strided convolutions (Decoder).
- Use BatchNorm in both the Encoder and the Decoder.
- Remove fully connected hidden layers.

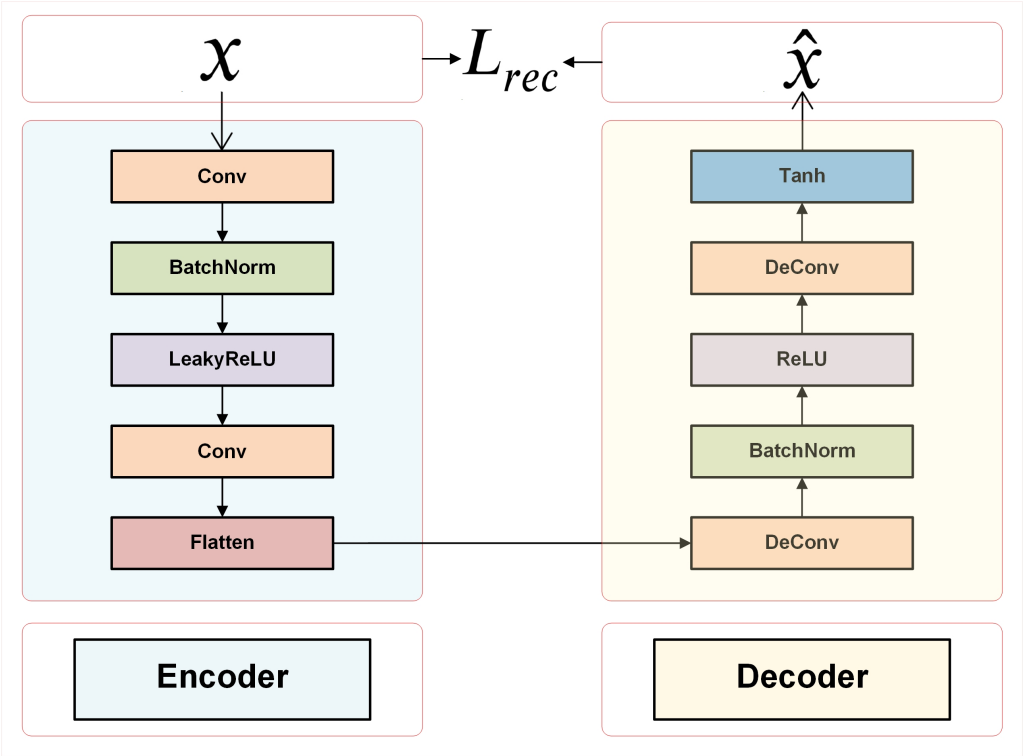


FIGURE 4 The architecture schematic diagram of the CAE

- Use ReLU activation in Decoder for all layers except for the output, which uses Tanh.
- Use LeakyReLU activation in the Encoder for all layers.

To search for better WANN, the high-dimensional latent representation needs to satisfy two conditions: (1) keeping the main and important characteristics of the raw data as much as possible; (2) more easily distinguished by the WANN than the raw data. In order to satisfy the above conditions, the objective function L_{cae} formulates by combining two-loss values (Reconstruction Loss and Feedback Loss), each of which has its own functions during the training stage.

Reconstruction Loss: In order to maximize the ability of the high-dimensional latent representation z to keeps the main and important characteristics of the raw data x, it can be achieved by minimizing the distance between the raw data x and the reconstructed representation $\hat{x}$. It is known that using L_1-distance yields less blurry results than L_2-distance[28], that is, using L_1-distance encourages $\hat{x}$ closer to x. Therefore, L_1-distance is used to measure the reconstruction loss between x and $\hat{x}$. The reconstruction loss is defined as below.

$$L_{rec} = \frac{1}{|X|} \sum_{x \in X} |x - \hat{x}|_1 \tag{2}$$

where X is the training set.

Feedback Loss: The reconstruction loss is adequate to guarantee that the obtained high-dimensional latent representation satisfies the first condition. However, with only a reconstruction loss, it is difficult to obtain the high-dimensional latent representation that satisfies the second condition. In fact, there is more than one high-dimensional latent representation of the input x. To obtain an easily distinguishable high-dimensional latent representation, the objective function L_{wann} is used, which is regarded as the feedback of the WANN. The feedback loss is defined as below.

$$L_{feed} = L_{wann} \tag{3}$$

Finally, the objective function of the CAE is as follows:

$$L_{cae} = L_{rec} + L_{feed} \tag{4}$$

2.4 | Weight Agnostic Neural Network Search

Weight agnostic neural network search is a novel network architecture search method that can accomplish specific tasks by only optimizing the architectures[21]. The main goal of this work is to find a suitable neural network architecture for classification tasks in search space[1].

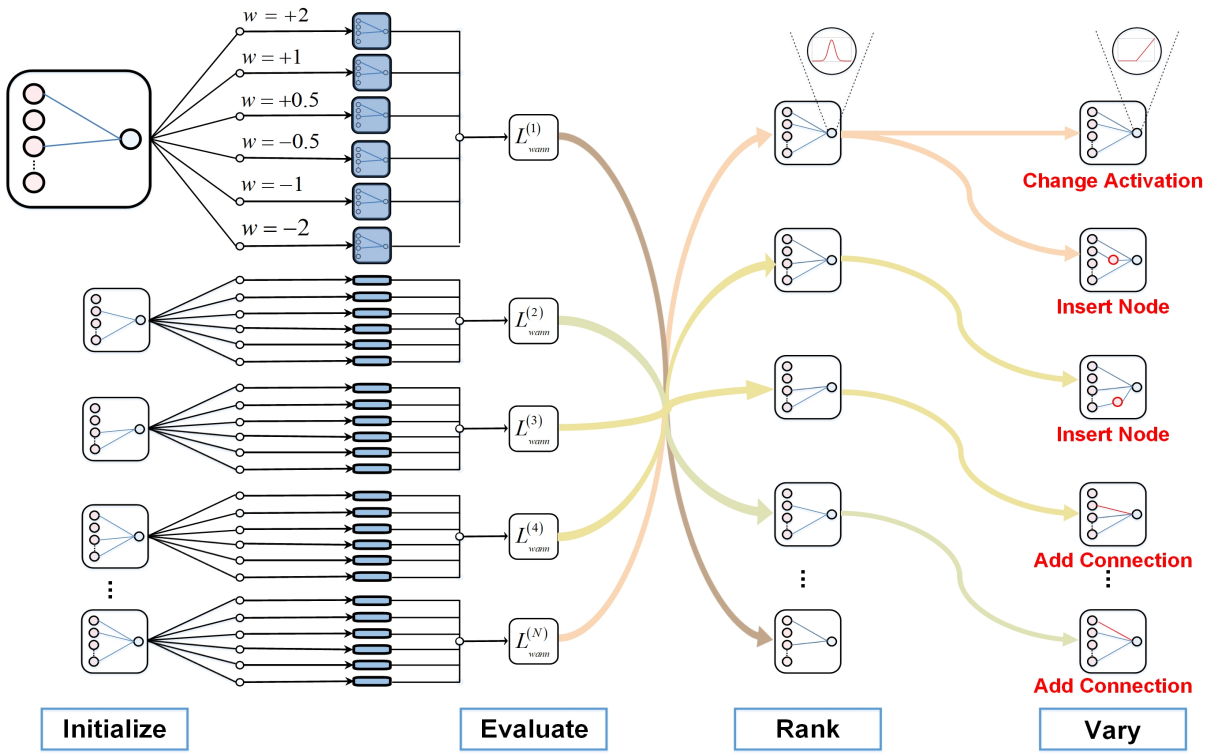


FIGURE 5 The evolution process of weight agnostic neural network search[21]

The search for WANNs can be summarized as follows(FIGURE 5): (1) Initialize: Different neural network architectures, called an initial population, are randomly created in the search space and only consist of the input layer and output layer. Population size is the number of such network architectures. (2) Evaluate: Each network architecture is assigned a series of different shared weights and evaluated by the objective function. In fact, the objective function includes two metrics. One is the performance metric used to avoid underfitting, the other is the complexity metric used to avoid overfitting. (3) Rank: All the network architectures are ranked according to their objective function. The higher the ranking of the network architecture is, the greater the selected probability is. (4) Vary: New network architectures are created by varying the selected network architectures. The operators for varying the selected network architectures include inserting node, adding connection, or changing activation, which are inspired by the NeuroEvolution of Augmenting Topologies(NEAT)[29]. More specific details are well described in the previous literature[1]. Then repeat from (2) to (4) until a suitable network architecture is searched.

2.4.1 | Objective Function To Guide The Neural Network Evolution

The network architecture evolution has to follow two different directions: (1) minimizing the error between the sigmoid output of the neural network and the corresponding label for avoiding underfitting; (2) reducing the number of connections in neural networks for avoiding overfitting. Therefore, the objective function L_{wann} consists of three-loss values (Mean Performance Loss, Max Performance Loss, and Complexity Loss), each of which has great influence in guiding the network architecture evolution[1].

Mean Performance Loss: The performance of the network architecture is evaluated by using the mean and max performance loss in the training phase. Additionally, the cross-entropy loss is used to minimize the error between the sigmoid output of the neural network and the corresponding label. Hence, the mean performance loss L_{mean} is defined as below.

$$L_{mean} = \frac{1}{|W|} \sum_{w \in W} \left\{ - \sum_{z \in Z} y_z \log(f(z,\ w)) + (1 - y_z) \log(1 - f(z,\ w)) \right\} \tag{5}$$

y_z represents the label of the latent representation z; $f(z, w)$ denotes the sigmoid output of the neural network that is assigned a shared weight w; Z is a set of the latent representation; and W is a set of different shared weight values. According to the previous reference[21], the shared weight values of this paper are as follows: $[-2, -1, -0.5, +0.5, +1, +2]$.

Max Performance Loss: Compared with the mean performance loss, the max performance loss L_{max} pays more attention to the potentiality of the network architecture, which is defined as below.

$$L_{max} = \min_{w \in W} \left\{ - \sum_{z \in Z} y_z \log(f(z,\ w)) + (1 - y_z) \log(1 - f(z,\ w)) \right\} \tag{6}$$

where $|W|$ denotes the number of shared weight values in W.

Complexity Loss:The mean and max performance loss can ensure that the network architecture evolution follows the first direction. However, with only the mean and max performance loss, the network architecture is easy to become very complex and fall into overfitting. It requires the complexity loss to guide the network architecture evolution. Additionally, its complexity is denoted by the number of connections in the network architecture. The complexity loss L_{comp} is defined as below.

$$L_{comp} = - \frac{1}{N_{connection}} \tag{7}$$

where $N_{connection}$ is the number of connections in the network architecture.

However, the network architecture is difficult to change greatly under a strict complexity constraint, which becomes the bottleneck of the performance increase. In order to balance this contradiction, the network architecture is probabilistically evaluated by complexity[1]. Each evaluation depends on a random variable r that obeys a Bernoulli distribution and takes the value one with probability 0.8. The network architecture is evaluated by using the mean performance loss and complexity loss when the value of the random variable equals one. Otherwise, the evaluation is done by the mean performance loss and max performance loss[21]. The objective function of the weight agnostic neural network is defined as follows:

$$L_{wann} = \begin{cases} L_{max} + L_{mean}\ , & r \leq 0.2 \\ L_{mean} + L_{comp}\ , & r > 0.2 \end{cases} \tag{8}$$

2.5 | Threshold Selection

In practice, only small amounts of abnormal data can be gathered from an airline fleet, and the number of normal samples is far more than that of abnormal samples. In other words, the monitoring data of gas turbines is unbalanced. Undoubtedly, WANNs pay less attention to abnormal samples during the training stage, so their sigmoid outputs are affected to a certain extent. As a footnote, the sigmoid output of the WANN is defined as the anomaly score of the sample. In this case, if the threshold is still set to 0.5, it may lead to poor diagnostic results. As Provost discussed that "the bottom line is that when studying problems with imbalanced data, using the classifiers produced by standard machine learning algorithms without adjusting the output threshold may well be a critical mistake"[30]. Therefore, it is not easy to obtain a suitable threshold for the WANN in the case of unbalanced datasets[1].

The commonly used threshold selection methods include the empirical method, the histogram method, the adaptive decision threshold(ADT), and so on[31 32 33]. The empirical method requires relevant professional knowledge, and the histogram method requires significant differences between the two categories. Moreover, the adaptive decision method only considers the ratio between the normal sample size and the abnormal sample size and ignores the distribution information of their anomaly scores. Since the considered information is not complete, the samples are also more easily misjudged.

To alleviate this problem, a novel method based on maximum likelihood estimation (MLETS) is proposed for the threshold selection. Now assume that the anomaly scores of normal and abnormal samples approximately obey the truncated normal distribution. The probability density function(PDF) of the anomaly score s_n of the normal sample is defined as below.

$$\varphi_1\left(s_n;\ \mu_1,\ \sigma_1,\ a_1,\ b_1\right) = \begin{cases} 0 & ,\ s_n < a_1 \\ \frac{\phi(s_n;\ \mu_1,\ \sigma_1)}{\Phi(b_1;\ \mu_1,\ \sigma_1) - \Phi(a_1;\ \mu_1,\ \sigma_1)} & ,\ a_1 \leq s_n \leq b_1 \\ 0 & ,\ s_n > b_1 \end{cases} \tag{9}$$

where μ_1 and σ_1 are the mean and variance of the truncated normal distribution φ_1; a_1 and b_1 specify the truncation interval; ϕ and Φ are the PDF and cumulative distribution function(CDF) of the normal distribution. Note that $0 \leq s_n \leq 1$, then $a_1 = 0$, $b_1 = 1$. To estimate the parameters $\hat{\mu}_1$ and $\hat{\sigma}_1$, this can be achieved by maximizing the log-likelihood function of the normal samples in the training set, which is called the maximum likelihood estimate. And the log-likelihood function is defined as below.

$$L\left(\mu_1, \sigma_1\right) = \sum \ln \varphi_1\left(s_n; \mu_1, \sigma_1, 0, 1\right) \tag{10}$$

Then the maximum likelihood estimation of the parameters $\hat{\mu}_1$,$\hat{\sigma}_1$ is as follows:

$$\hat{\mu_1},\ \hat{\sigma_1} = \max_{\mu_1, \sigma_1} \left\{L\left(\mu_1, \sigma_1\right)\right\} \tag{11}$$

Then the PDF of the anomaly score s_a of the abnormal sample is defined as below.

$$\varphi_2\left(s_a;\ \mu_2,\ \sigma_2,\ a_2,\ b_2\right) = \begin{cases} 0 & ,\ s_a < a_2 \\ \frac{\phi\left(s_a;\ \mu_2,\ \sigma_2\right)}{\Phi\left(b_2;\ \mu_2,\ \sigma_2\right) - \Phi\left(a_2;\ \mu_2,\ \sigma_2\right)} & ,\ a_2 \leq s_a \leq b_2 \\ 0 & ,\ s_a > b_2 \end{cases} \tag{12}$$

where μ_2 and σ_2 are the mean and variance of the truncated normal distribution φ_2; a_2 and b_2 specify the truncation interval. Note that $0 \leq s_a \leq 1$, then $a_2 = 0$, $b_2 = 1$. Similarly, the log-likelihood function of the abnormal samples in the training set is defined as below.

$$L\left(\mu_2, \sigma_2\right) = \sum \ln \varphi_2\left(s_a; \mu_2, \sigma_2, 0, 1\right) \tag{13}$$

And the maximum likelihood estimation of the parameters μ_2,σ_2 is as follows:

$$\hat{\mu_2},\ \hat{\sigma_2} = \max_{\mu_2, \sigma_2} \left\{L\left(\mu_2, \sigma_2\right)\right\} \tag{14}$$

According to Equ.11 and Equ.14, the maximum likelihood estimation of the parameters $\hat{\mu}_1,\ \hat{\mu}_2,\ \hat{\sigma}_1,\ \hat{\sigma}_2$ can be obtained. And the log-likelihood function of the training set is defined as below.

$$L\left(\beta\right) = \sum_{s \in S} \ln\left[\varphi_1\left(s;\ \hat{\mu}_1,\ \hat{\sigma}_1,\ 0,\ 1\right) I\left\{s < \beta\right\} + \varphi_2\left(s;\ \hat{\mu}_2,\ \hat{\sigma}_2,\ 0,\ 1\right) I\left\{s \geq \beta\right\}\right] \tag{15}$$

where β is the decision threshold of the classifier and $I\left\{\cdot\right\}$ represents the characteristic function.

Then the maximum likelihood estimation of β is as follows:

$$\hat{\beta} = \max_{\beta \in [0,\ 1]} \left\{L\left(\beta\right)\right\} \tag{16}$$

And $\hat{\beta}$ is set as the decision threshold of the WANN. More concretely, the gas turbine is in abnormal condition when its anomaly score is not less than the threshold. Otherwise, it is in normal condition.

3 | EXPERIMENTS AND RESULTS

3.1 | Data Set Preparation And Preprocessing

In this paper, actual data of gas turbines is collected to evaluate the proposed anomaly detection model, which comes from the real-life operation data of an Asia airline rather than simulated data. Theoretically, some performance parameters deviate from the original trend when gas turbines fail. Among them, Delta Exhaust Gas Temperature(DEGT), Delta Core Speed(DN2), and Delta Fuel Flow(DFF) have more direct and obvious changes, which are used for fault diagnosis by the Original Equipment Manufacturer(OEM) as well. Therefore, DEGE, DN2, and DFF are chosen for anomaly detection. Besides, we collected maintenance reports of the airline from 2008 to 2015. The removal date can be determined according to maintenance reports, and the status of the gas turbine can be judged as well. Then samples are constructed using sliding window method. According to the relevant Customer Notification Report(CNR), it is seen that the length of the abnormal sequence is more than five, and the vast majority is not more than ten. Therefore, the size of the sliding window is finally set to ten. Moreover, the collected samples are summarized in TABLE 1.

TABLE 1 Gas turbine dataset

No	DEGT	DN2	DFF	State
1	30.6606	-0.2946	0.6565	normal
	22.4841	-0.6491	0.234	
	20.3759	-0.7301	-0.657	
	19.9201	-0.0157	1.4827	
	22.3744	-0.8221	-0.408	
	20.2971	-0.6535	-0.2862	
	21.6608	-0.9676	0.2019	
	21.9958	-0.814	0.415	
	27.3981	-0.471	0.0942	
	24.8495	-0.6588	0.5205	
	...	...	...	
512	28.0694	-0.9336	2.3797	abnormal
	23.4776	-0.8427	1.4983	
	13.1551	-1.2589	-0.0978	
	39.0245	-0.8254	4.3871	
	35.0468	-0.716	2.1861	
	17.7177	-1.1165	0.7452	
	16.949	-0.9617	0.38	
	16.2494	-1.194	-0.6943	
	22.766	-1.0596	2.0265	
	15.8263	-1.1538	-0.0533	

3.2 | Experimental Setup

The choice of hyperparameters is introduced clearly as follows. The WANN-related hyperparameters are used to define population size, generations, and the probability of three operations during the search. Hyperparameters (all except generations) are set according to the MNIST classification experiment[21], as shown in TABLE 2. The number of generations is set to 512 according to trial tests. FIGURE 6(a) shows that the mean performance loss and max performance loss of the best WANN almost converge after 512 generations.

TABLE 2 Hyperparameters of the WANN

Hyperparameters I	MNIST[21]	Our
Population Size	960	960
Generations	**4096**	**512**
Change Activation Probability (%)	50	50
Insert Node Probability (%)	25	25
Add Connection Probability (%)	25	25
Initial Active Connections Probability (%)	5	5
Tournament Size	32	32

The CAE-related hyperparameters are used to define the structure of the neural networks, including channels of the convolution, size of the convolution kernel, and stride of the convolution, and so on. First of all, the size of the convolution kernel in the first convolution layer is set to 2*2, and that of the convolution kernel in the second convolution layer is set to 3*3 according to

the recommendations of reference [18]. Second, the stride of the convolution and zero-padding added to both sides are used with its default settings. Third, the channels of the convolution are set according to trial tests. FIGURE 6(b) shows that the reconstruction loss of the CAE almost converges to a similar value under different channels conditions. Therefore, the channels of the first convolution layer are set to 2, and that of the second convolution layer are set to 4. The hyperparameters of the CAE are summarized in TABLE 3.

TABLE 3 Hyperparameters of the CAE

Hyperparameters II	Conv1	Conv2	DeConv1	DeConv2
Channels of the convolution	2	4	4	2
Size of the convolution kernel	2*2	3*3	3*3	2*2
Stride of the convolution	1	1	1	1
Zero-padding added to both sides	1	0	0	1

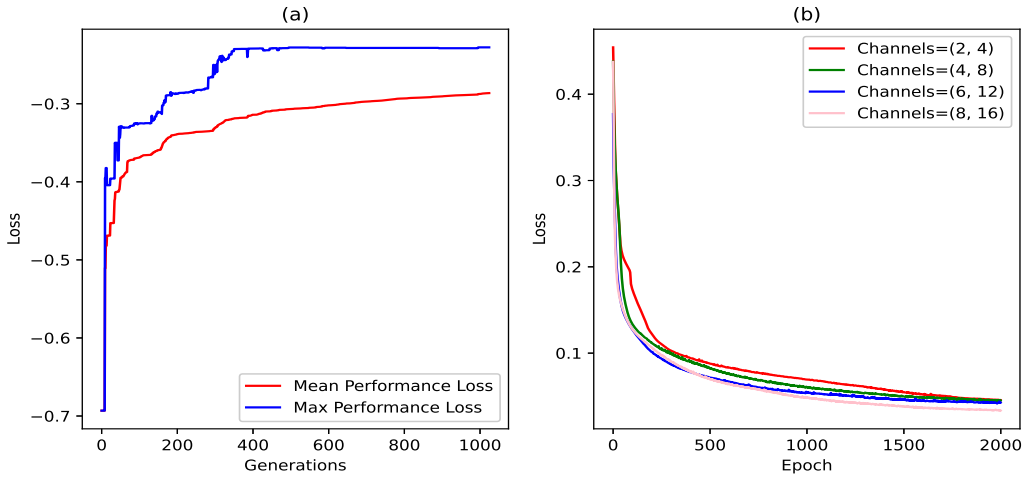


FIGURE 6 (a) The mean performance loss and max performance loss of the best WANN (b) the reconstruction loss of the CAE under different channels conditions

3.3 | Evaluation Metrics

In fact, gas turbines have a low failure frequency due to their high reliability, and the number of normal samples is far more than that of abnormal samples. That is, the dataset is unbalanced, which is summarized in TABLE 4. In this case, the overall accuracy has no significant guidance, because it is mainly determined by normal samples. Now that the number of normal samples in the test set is 130, and that of abnormal samples is 25. Even if abnormal samples are all diagnosed as normal samples, the overall accuracy is about 0.84. Although the overall accuracy is not too low, it has no important reference value. Therefore, the AUC and confusion matrix are used to validate the proposed model in this paper.

TABLE 4 Sample size

State	Training	Testing	Total
Normal	301	130	431
Abnormal	56	25	81
Total	357	155	512

3.3.1 | AUC

The area under the receiver operating characteristic (ROC) curve, called AUC, is statistically consistent. And studies have shown that AUC is a more discriminating measure than accuracy[34]. What's more, it is also capable of validating the performance of the anomaly detection model even if the dataset is unbalanced.

3.3.2 | Confusion Matrix

According to the confusion matrix(TABLE 5), it is easy to calculate the true positive rate (TPR), the true negative rate (TNR), and the Youden index (YI), which are defined as below. The larger their values are, the better the anomaly detection model is.

$$TPR = \frac{TP}{TP + FN} \tag{17}$$

$$TNR = \frac{TN}{TN + FP} \tag{18}$$

$$YI = TPR + TNR - 1 \tag{19}$$

TABLE 5 An illustration of confusion matrix

Actual Class	**Predicted Class**	
	Positive	**Negative**
Positive	True Positive (TP)	False Negative (FN)
Negative	False Positive (FP)	True Negative (TN)

3.4 | Experimental Results

3.4.1 | The Anti-noise Performance of the CAE in Obtaining the Latent Representation of the Raw Data

The purpose of establishing CAE is to obtain the latent representation of the raw data while keeping the main information and enhancing the separability between abnormal and normal samples. The raw data is transformed into 1D vector for display purposes. The dimension represents the position of the element in the vector and the value represents the numerical value of the element. FIGURE 7 shows the changing trend of normal samples in different spaces, while FIGURE 8 shows the changing trend of abnormal samples in different spaces.

Combined with FIGURE 7 and FIGURE 8, it is obvious that the samples of the same category in the high-dimensional space are better clustered together compared with the original space. Furthermore, using the CAE to map raw data to the high-dimensional space can effectively suppress the classification interference caused by data noise. Hence, the CAE can enhance the separability between abnormal and normal samples to some extent.

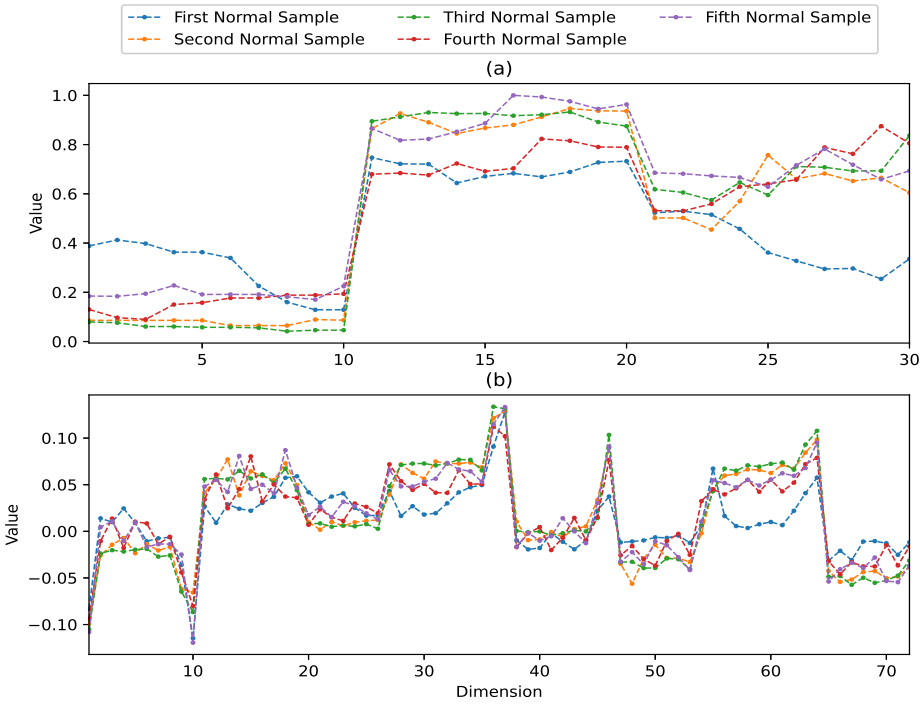


FIGURE 7 The raw data and high-dimensional latent representation of the five normal samples

3.4.2 | The Effectiveness of the Proposed CAE-WANN

To validate the effectiveness of the CAE-WANN, this paper establishes five groups of contrast experiments. In the first group of experiments, the raw data is directly classified by the support vector machine(SVM). The strength of the regularization is set to be one hundred, and the selected kernel types include the common linear and polynomial kernel functions. They are separately recorded as the SVM(Linear) and SVM(Ploy). In the second group of experiments, the raw data is directly diagnosed by the decision tree(DT). The strategy used to choose the split at each node is the best split, and the criteria to measure the quality of a split include the common the Gini impurity and the information gain. They are separately recorded as the DT(Gini) and DT(Entropy). In the third group of experiments, the raw data is directly detected by BP neural networks(BPNN) with four layers. The input layer has thirty nodes, while the output layer has one node. And the first hidden layer has twenty nodes, while the second hidden layer has ten nodes. Under small sample size, it is difficult to select the number of nodes in the hidden layer. In the fourth group of experiments, the raw data is directly classified by the WANN. In the fifth group of experiments, the CAE is used to obtain the high-dimensional latent representation of raw data, and then the high-dimensional latent representation is classified by weight agnostic neural network. Its hyperparameters are also set according to TABLE 2 and TABLE 3. For ease of differentiation, this contrast experiment is recorded as the CAE-WANN(separately). In addition, each experiment is repeated ten times for eliminating the randomness of neural networks.

TABLE 6 shows the AUC mean and standard deviation of the SVM, DT, BPNN, and WANN with different shared weight values. Compared with the conventional machine learning methods such as SVM, DT, and BPNN, the WANN obtained better diagnostic results in each case. More specifically, compared with the SVM(Linear), SVM(Poly), DT(Gini), DT(Entropy), and BPNN, the AUC mean of the WANN with the best shared weight values -1 is improved by about 295.80%, 52.21%, 12.05%, 15.57%, and 6.43%. Even for the worst shared weight values -0.5, the AUC mean is improved by about 278.23%, 45.45%, 7.07%, 10.44%, and 1.70%. It's worth noting that the number of connections in the BPNN is 810, and that in the WANN is approximately 160. Obviously, the WANN is not only better but also smaller than the BPNN. Additionally, TABLE 6 also shows the AUC mean and standard deviation of the CAE-WANN(Separately) and CAE-WANN with different shared weight values. In most cases, the CAE-WANN has a higher AUC mean than the WANN and CAE-WANN(Separately). In addition, the AUC standard deviation of the CAE-WANN is smaller than that of the WANN and CAE-WANN(Separately). Therefore, the CAE-WANN not only searches for a more stable model but also yields more excellent performance than the WANN and CAE-WANN(Separately).

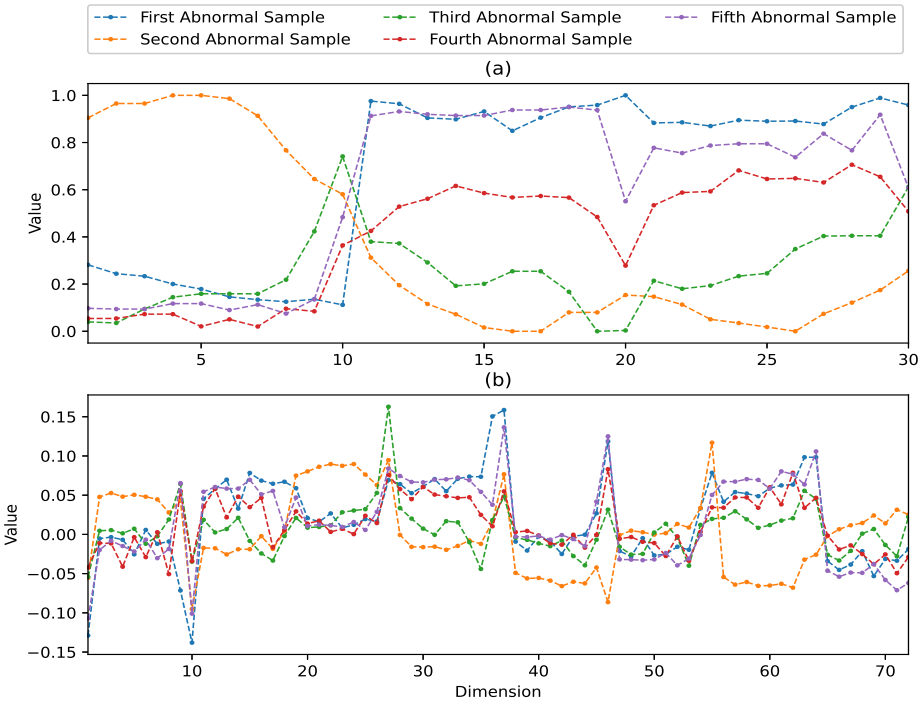


FIGURE 8 The raw data and high-dimensional latent representation of the five abnormal samples

TABLE 6 shows the inference time performance of the different anomaly detection models on the test set. The DT(Gini) and DT(Entropy) have the best inference time and take roughly 0.07 milliseconds to diagnose one hundred and fifty-five test samples, while the proposed CAE-WANN takes over 2.78 milliseconds. The difference between the CAE-WANN and DT is about 2.71 milliseconds, which can be acceptable in engineering practice. Moreover, compared with the DT(Gini) and DT(Entropy), the AUC mean of the CAE-WANN with the best shared weight values 1 is improved by about 14.48% and 18.08%, which significantly improves the anomaly detection performance. Even for the worst shared weight values 2, the AUC mean is improved by about 6.19% and 9.53%. On the basis of comprehensive consideration, the proposed anomaly detection method is more potential for gas turbine anomaly detection in engineering practice.

3.4.3 | The Effectiveness of the Proposed Threshold Telection Method

To verify the effectiveness of the proposed MLETS, this paper establishes two groups of comparison experiments. In the first group of experiments, the fixed decision threshold(FDT) is used to diagnose the anomaly scores. And the ADT is selected in the second group of experiments. But before it, the Kolmogorov-Smirnov test provided by SciPy is selected to demonstrate that the anomaly scores obey truncated normal distributions[35]. If the p-value of the Kolmogorov-Smirnov test is greater than the significance level α(Default 0.05), the random variable obeys the probability distribution. For normal samples, the p-value is 0.9882, which is greater than the significance level 0.05. For abnormal samples, the p-value is 0.9807, which is also greater than the significance level 0.05. Therefore, the anomaly scores of normal and abnormal samples approximately obey the truncated normal distribution.

The optimal thresholds are calculated according to Equ.16, which is shown in TABLE 7. It is obvious that CAE-WANN with different shared weight values has various diagnostic capabilities. Inspired by ensemble learning, the naive voting ensemble approach is used to combine multiple diagnostic results to form a better final diagnostic result in this paper. Each of WANNs with different shared weight values is granted equal voting rights and has only one choice between the normal and abnormal sample. If the number of abnormal votes is not less than two, it is diagnosed as an abnormal sample.

TABLE 8 provides a straightforward comparison of using individual models and the voting approach. This voting approach yields predictions far better than individual models in terms of TNR and YI, and only slightly worse than them in terms of

TABLE 6 AUC mean, standard deviation, and average inference time of the different anomaly detection models

Method		Mean(%)	Standard Deviation(%)	Inference Time(milliseconds)
SVM(Linear)		21.45	**0**	0.51
SVM(Poly)		55.78	**0**	0.44
DT(Entropy)		73.46	**0**	**0.07**
DT(Gini)		75.77	**0**	**0.07**
BPNN		79.77	2.25	0.12
WANN	-2	82.36	3.04	1.26
	-1	84.90	3.61	1.31
	-0.5	81.13	5.61	1.33
	+0.5	82.82	6.11	1.36
	+1	81.34	8.27	1.33
	+2	81.98	4.12	1.33
CAE-WANN (Separately)	-2	78.39	4.36	2.71
	-1	82.54	3.60	2.76
	-0.5	82.41	2.28	2.80
	+0.5	82.86	4.40	2.76
	+1	83.40	2.09	2.75
	+2	78.39	4.95	2.74
CAE-WANN	-2	82.51	3.42	2.80
	-1	86.27	2.82	2.78
	-0.5	85.25	3.20	2.79
	+0.5	85.67	3.00	2.81
	+1	**86.74**	1.69	2.81
	+2	80.46	2.67	2.82

TPR. It is evident that the voting approach can improve the balanced diagnostic capability of the MLETS. Moreover, the final diagnostic result is summarized in TABLE 9. There were 130 normal samples, of which 119 normal samples are correctly diagnosed, so the TPR of the FDT is about 91.54%. Besides, 18 abnormal samples are correctly diagnosed, and the TNR of the FDT is 72.00%. The YI of the FDT is about 63.45%. The TPR of the ADT is about 72.31%, and its TNR is 92.00%. The YI of the ADT is about 64.31%. Compared with the FDT and ADT, the results show that the proposed MLETS has the most balanced diagnostic capability. Therefore, the MLETS has great application potential in engineering practice.

TABLE 7 Confusion matrix of the CAE-WANN with different weight values

Shared Weight	Threshold	Actual Class	Predicted Class	
			Normal	Abnormal
-2	**0.4218**	**Normal**	119	11
		Abnormal	10	15
-1	**0.2706**	**Normal**	115	15
		Abnormal	7	18
-0.5	**0.2886**	**Normal**	116	14
		Abnormal	8	17
0.5	**0.2680**	**Normal**	115	15
		Abnormal	7	18
1	**0.2729**	**Normal**	116	14
		Abnormal	10	15
2	**0.4209**	**Normal**	119	11
		Abnormal	11	14

TABLE 8 TPR, TNR, and YI of the proposed anomaly detection framework

Individual Models vs Voting		TPR(%)	TNR(%)	YI (%)
Individual Models	**-2**	**91.54**	60.00	51.54
	-1	88.46	72.00	60.46
	-0.5	89.23	68.00	57.23
	0.5	88.46	72.00	60.46
	1	89.23	60.00	49.23
	2	**91.54**	56.00	47.54
Voting		85.38	**84.00**	**69.38**

4 | CONCLUSION

In this paper, an anomaly detection method for gas turbines is presented by combining the WANN and CAE. The objective function of the WANN is used to avoid overfitting or underfitting. On one hand, the mean and max performance loss are capable of minimizing the error between the output of the neural network and the corresponding label for avoiding underfitting. On the other hand, complexity loss is capable of reducing the number of connections in neural networks for avoiding overfitting. And the CAE is used to extend the search space, which improves not only the detection performance of the WANN but also the stability of diagnostic results. Moreover, a novel method based on maximum likelihood estimation is proposed for threshold selection in the case of unbalanced datasets. The experimental results show that the combination of the WANN and CAE can not only obtain a better diagnostic result but also have better stability than the conventional BPNN and WANN. As a result, the presented anomaly detection model has a higher application value for gas turbines. Compared with previous search algorithms for neural

TABLE 9 Final confusion matrix of the WANN

Method	Threshold	Actual Class	Predicted Class	
			Normal	Abnormal
FDT[31]	**0.5**	**Normal**	119	11
		Abnormal	7	18
ADT[31]	**0.3016**	**Normal**	94	36
		Abnormal	2	23
proposed MLETS	——	**Normal**	111	19
		Abnormal	4	21

network architectures, it directly evaluates architectures without optimizing the weights, which greatly accelerates the speed of the network search. However, the evaluation for architectures still makes the search costly. In addition, the naive voting ensemble approach ignores the diagnostic ability of WANNs with different shared weight values, which may have a certain negative impact on the accuracy of the final diagnosis. Therefore, our future efforts will concentrate on the discovery of an effective acceleration method that can greatly save search time and a new ensemble approach that may lead to better diagnostic results.

ACKNOWLEDGMENTS

Thanks to the National Natural Science Foundation of China (No.U1733201) for providing support for this paper.

References

1. Zhong S, Liu D, Lin L, Zhao M, Fu X, Guo F. A novel anomaly detection method for gas turbines using weight agnostic neural network search. In: IEEE. 2020 Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling (APARM); 2020: 1–6.

2. Tayarani-Bathaie SS, Vanini ZS, Khorasani K. Dynamic neural network-based fault diagnosis of gas turbine engines. *Neurocomputing* 2014; 125: 153–165.

3. Yousefi N, Coit DW, Song S, Feng Q. Optimization of on-condition thresholds for a system of degrading components with competing dependent failure processes. *Reliability Engineering System Safety* 2019; 192: 106547.

4. Awad M. Fault detection of fuel systems using polynomial regression profile monitoring. *Quality and Reliability Engineering International* 2017; 33(4): 905–920.

5. Asgari H, Chen X, Sainudiin R. Modelling and simulation of gas turbines. *International Journal of Modelling, Identification and Control* 2013; 20(3): 253–270.

6. Yu L, Cleary DJ, Cuddihy PE. A novel approach to aircraft engine anomaly detection and diagnostics. In: IEEE. 2004 IEEE Aerospace Conference Proceedings; 2004: 3468–3475.

7. Tian Z, Jin T, Wu B, Ding F. Condition based maintenance optimization for wind power generation systems under continuous monitoring. *Renewable Energy* 2011; 36(5): 1502–1509.

8. Bzdok D, Altman N, Krzywinski M. Points of significance: statistics versus machine learning. *Nature Methods 2018a* 2018: 1–7.

9. Wang FK, Mamo T. Hybrid approach for remaining useful life prediction of ball bearings. *Quality and Reliability Engineering International* 2019; 35(7): 2494–2505.

10. Fu X, Chen H, Zhang G, Tao T. A new point anomaly detection method about aeroengine based on deep learning. In: IEEE. 2018 International Conference on Sensing, Diagnostics, Prognostics, and Control (SDPC); 2018: 176–181.

11. Brotherton T, Johnson T. Anomaly detection for advanced military aircraft using neural networks. In: IEEE. 2001 IEEE Aerospace Conference Proceedings; 2001: 3113–3123.

12. Lee H, Li G, Rai A, Chattopadhyay A. Health monitoring framework for aircraft engine system using deep neural network. In: No. 1. Proceedings of the Annual Conference of the PHM Society; 2019.

13. Li Z, Zhong S, Lin L. Novel gas turbine fault diagnosis method based on performance deviation model. *Journal of Propulsion and Power* 2017; 33(3): 730–739.

14. Yuan M, Wu Y, Lin L. Fault diagnosis and remaining useful life estimation of aero engine using LSTM neural network. In: IEEE. 2016 IEEE International Conference on Aircraft Utility Systems (AUS); 2016: 135–140.

15. Xie L, Wang J, Wei Z, Wang M, Tian Q. Disturblabel: Regularizing cnn on the loss layer. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition; 2016: 4753–4762.

16. Merity S, Keskar NS, Socher R. Regularizing and optimizing LSTM language models. *arXiv:1708.02182* 2017.

17. Cogswell M, Ahmed F, Girshick R, Zitnick L, Batra D. Reducing overfitting in deep networks by decorrelating representations. *arXiv:1511.06068* 2015.

18. Zhong S, Fu S, Lin L. A novel gas turbine fault diagnosis method based on transfer learning with CNN. *Measurement* 2019; 137: 435–453.

19. So DR, Liang C, Le QV. The evolved transformer. *arXiv:1901.11117* 2019.

20. Zoph B, Le QV. Neural architecture search with reinforcement learning. *arXiv:1611.01578* 2016.

21. Gaier A, Ha D. Weight agnostic neural networks. In: Advances in Neural Information Processing Systems; 2019: 5364–5378.

22. Elsken T, Metzen JH, Hutter F. Neural architecture search: A survey. *arXiv:1808.05377* 2018.

23. Ranzato M, Huang FJ, Boureau YL, LeCun Y. Unsupervised learning of invariant feature hierarchies with applications to object recognition. In: IEEE. 2007 IEEE Conference on Computer Vision and Pattern Recognition; 2007: 1–8.

24. Masci J, Meier U, Cirean D, Schmidhuber J. Stacked convolutional auto-encoders for hierarchical feature extraction. In: Springer. International Conference on Artificial Neural Networks; 2011: 52–59.

25. Turchenko V, Luczak A. Creation of a deep convolutional auto-encoder in caffe. In: IEEE. 2017 9th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS); 2017: 651–659.

26. Robinson RW. Counting unlabeled acyclic digraphs. In: Combinatorial mathematics V. 1977 (pp. 28–43).

27. Radford A, Metz L, Chintala S. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv:1511.06434* 2015.

28. Isola P, Zhu JY, Zhou T, Efros AA. Image-to-image translation with conditional adversarial networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition; 2017: 1125–1134.

29. Stanley KO, Miikkulainen R. Evolving neural networks through augmenting topologies. *Evolutionary Computation* 2002; 10(2): 99–127.

30. Provost F. Machine learning from imbalanced data sets 101. In: AAAI Press. Proceedings of the AAAI 2000 Workshop on Imbalanced Data Sets; 2000: 1–3.

31. Wu K, Zhong S, Fu X, Wei C. An aeroengine gas path anomaly detection method in the case of sample imbalance. In: IEEE. 2019 Prognostics and System Health Management Conference (PHM-Qingdao); 2019: 1–6.

32. Zhou ZH, Liu XY. Training cost-sensitive neural networks with methods addressing the class imbalance problem. *IEEE Transactions on Knowledge and Data Engineering* 2005; 18(1): 63–77.

33. Collell G, Prelec D, Patil KR. A simple plug-in bagging ensemble based on threshold-moving for classifying binary and multiclass imbalanced data. *Neurocomputing* 2018; 275: 330–340.

34. Ling CX, Huang J, Zhang H. AUC: a statistically consistent and more discriminating measure than accuracy. In: Ijcai; 2003: 519–524.

35. Massey FJ. The Kolmogorov-Smirnov test for goodness of fit. *Journal of the American Statistical Association* 1951; 46(253): 68–78.

36. Liu D, Zhong S, Lin L, et al. A Novel Performance Prediction Method for Gas Turbines Using the Prophet Model. In: IEEE. 2021 IEEE International Conference on Sensing, Diagnostics, Prognostics, and Control (SDPC); 2021: 203–208.

